

Platform

```
npm install
npm run demo          # the data side: apps, installs, permissions, receipts
npm run demo:device   # the execution side: android, fan-out, repair queue, autom
npm run demo:modular  # slots, provider swap, model routing, harness, builder, im
npm run demo:surfaces # two MCP surfaces, the links page, availability + intent c
npm start             # HTTP API on :3000
```

`DATABASE_URL` switches it to real Postgres. Schema is `db/001_kernel.sql` ,
`db/002_workspace.sql` and `db/003_providers.sql` .

The split

Kernel is `src/` plus the two SQL files. It contains no business features. Everything else is a package in `modules/` , and every one of them is removable.

Packages come in kinds. Three matter so far:

- **app** — has its own data, a dashboard, and optionally a public section. QR menu, forms, CRM.
- **channel_app** — a real Android app the business installs on their own device and logs into. Facebook, Google Business Profile, Yelp. Carries an appmap.
- **automations** — data, not code. A trigger and a list of capability calls. Publishable.

Execution: the device, not an API

The executor is the business's own cloud Android instance. `provision()` creates two workloads on one host — the Android instance and a container — each with a persistent volume. The business installs real apps onto the Android instance and logs into them with their own accounts. The credential never leaves the workspace.

Each channel app ships an **appmap**: which canonical keys it carries, and for each key, the steps to write it and the steps to read it back.

```
"carries": ["hours", "contact.phone"],
```

```

"routes": {
  "hours": {
    "write": [{ "open": "com.facebook.katana" }, { "tap": "Hours" },
              { "type": "{{value.opens}}", "into": "opens_at" }, { "tap": "Save"
    "read":  [{ "open": "com.facebook.katana" }, { "tap": "Hours" },
              { "read": "opens_at" }, { "read": "closes_at" } ]},
    "assemble": { "opens": "opens_at", "closes": "closes_at" }
  }
}

```

`src/drivers/android.js` is the seam. Steps go in; a simulator runs them today, an ADB-backed cloud instance runs them later. Nothing above that file changes.

Fan-out: nobody says “update Facebook”

The owner changes hours once. `canonical.fact_changed` fires. `fanout.js` looks up every app installed on that device whose appmap declares it carries `hours`, and queues a `channel.push` against each one. The owner never picks destinations.

Real output from `demo-device.js`:

```

OWNER CHANGES HOURS ONCE, IN ONE PLACE
owner set hours to 07:00 - 22:00. nobody told the platform where to put it.

facebook          route android succeeded verification: verified
google_business   route android succeeded verification: verified
yelp               route android succeeded verification: verified

```

Verification is a screen read

A push is not believed because the steps ran. The verifier replays the appmap’s `read` steps, pulls the value off the screen, records it as that app’s own observation with an authority rank, and compares it to canonical. That comparison sets `in_sync`. A push that ran but did not land comes back `partial`, not `verified`.

When a map goes stale

Apps move things. When a step cannot find its target the run stops, writes a `repair_item` with the failing step and a screenshot, and fails. It does not guess at a nearby field and it does not report success:

```

YELP MOVES A FIELD (the map no longer matches the app)

```

```
result: failed, verification: failed
  repair queue: yelp — no field "from" on screen
    failing step: {"type":"07:00","into":"from"}
```

The live screen

POST /api/b/:slug/device/session opens a view or control session against the same instance the automations drive. Not a copy. GET /api/b/:slug/device/screen returns what is on it now.

Shared automations

One business writes an automation and publishes it. Another installs it by key — it holds a reference, not a copy. When the author publishes a new version, everyone who installed it gets the new behaviour without doing anything:

```
reef-charters installed "lead-to-crm" authored by ocean-view
  reef-charters contact: Cam Ruiz
after ocean-view published 1.1.0, with no action by reef-charters:
  deal: Inquiry from Priya Nandan (new)
```

Private stays private

Every business gets its own MCP surface at /mcp/:slug, built from the same projections that make the public profile. A package that did not declare a public section cannot appear on it. Its provisioned tables exist, are reachable by that app through the gateway, and are not on the surface.

Every layer is a slot

The kernel declares slots. A slot has a contract — the functions a provider must export. A provider package fills it. The kernel names no vendor anywhere.

workspace.executor	needs ["run", "screenshot", "prepare"]	one
model	needs ["models", "complete"]	many
harness	needs ["run"]	many
builder	needs ["plan"]	one
memory	needs ["remember", "recall"]	one
sandbox	needs ["test"]	one

Swapping is one row in `provider_binding`. A business binding shadows the platform default. From `demo-modular.js`:

SWAPPING THE EXECUTION LAYER

```
cloud executor: hours pushed, result {"packageKey":"facebook","key":"hours",...}
rebound to: android_local_node  ({"latency":"lan","location":"on_premise"})
pushes so far: verified, verified
same appmaps, same capabilities, same receipts. one row changed where it runs.
```

The contract is checked when the package loads, not at 3am — a provider that does not export what its slot requires is rejected at boot.

Model routing

Model providers declare what they have and what each one costs. The router scores candidates against the task and falls back when one fails, recording that it did:

```
need=reasoning          -> hosted_models:reasoning-large
need=cheap              -> hosted_models:fast-small
need=vision             -> local_models:node-vision
need=reasoning local-only -> local_models:node-small

with the hosted provider unreachable:
need=reasoning          -> local_models:node-small
recorded: chose local_models:node-small, fell back from hosted_models:fast-small
```

`local: true` on a task means the data may not leave the workspace, and only providers whose models declare `local` are eligible.

A harness cannot widen its own reach

A harness is handed `invoke`, which goes through the same gate, approvals and receipts as a dashboard click. It is offered only capabilities the acting person could already invoke:

```
capabilities offered to the owner's harness: 8
capabilities offered to the staff harness: 1
withheld from staff: business.set_fact, business.set_hours, channel.push,
                      channel.read, business_profile.update_contact,
                      qr_menu.set_availability, qr_menu.remove_item
```

Builder: an intent becomes a package

```
intent: build a public song request board customers can scan and add a song titl
proposed package: public_song_request (app)
table: public_song_request fields ["name","song","title","artist"]
capabilities: .create .list .update .remove
state: proposed — nothing exists yet
```

```
approved. package public_song_request installed.
signature: valid, signer business:ocean-view, tier community
created a row through the new capability: verification verified
public profile sections now: About | Hours | Menu | Public Song Request | Contac
```

A manifest with a schema gets create/list/update/remove and a renderer for free (`src/kernel/generated.js`), each with a real verifier. The generated package goes through the same validator and the same installer as a hand-written one.

Import: open source becomes a package or it stops

Eleven steps, and the job row records which one failed and why:

```
fetch -> analyze -> graph -> license -> security -> pin
      -> sandbox -> capabilities -> permissions -> sign -> publish

example/waitlist -> published as waitlist
  license MIT, pinned 9f2clab, signed
example/copyleft -> rejected at license: AGPL-3.0 cannot be redistributed inside
example/greedy    -> rejected at permissions: declares capability outside its nam
```

A branch is not a package — the commit is pinned. Nothing installs unsigned; the signature covers the manifest hash, so changing what a package declares invalidates it.

Scripts edit. Models read.

A capability that writes into an outside app through the device is marked `agentSafe: false`. The harness is never offered it, and the private surface labels it `scriptOnly`. The appmap is the whole decision — no model chooses which button to press when something is being changed.

```
capabilities offered to the owner's harness: 7
withheld: channel.push (script-only)
script-only on the owner surface: channel.push
```

`channel.read` is `agentSafe: true`. A model may open an app and read it, because reading cannot change anything.

Two surfaces per business

```
ghost:circle-boats (public) no credential
ghost://circle-boats/about from business_profile
ghost://circle-boats/hours from business_profile
ghost://circle-boats/availability from availability
ghost://circle-boats/menu from qr_menu
tools: read_about, read_hours, read_availability, read_menu, check_availability
```

```
ghost:circle-boats:private token bound to a membership
without a token: unauthorized
owner token -> acting as owner, 10 tools
staff token -> acting as staff, 2 tools
staff sees: availability.search, availability.hold (asks)
```

The private surface is not a fixed tool list. It is built per token from what that membership is granted, so a tool that would be denied never appears, and a tool whose disposition is `ask` is labelled as one that will park for approval. `POST /api/b/:slug/tokens` issues one; it is shown once and stored hashed.

The links page

`GET /l/:slug` is a real page. Every tile is a projection from an installed app, and every one opens in place:

```
sections rendered: Hours | Check Availability | Menu
page size: 6.5 kb, zero outbound links to a booking platform
```

Install an app, a tile appears. Uninstall it, the tile goes. The open/closed badge is computed from canonical hours with any temporary closure layered on top — it is not typed into the page.

`GET /embed/:slug/:section` returns the same section as a fragment for the business's own website. One source, three places: links page, embed, MCP.

The search is the asset

Because the visitor never leaves, the business keeps what a booking platform would have

kept:

```
what the business now knows:
links    2026-08-25  party 2   2 matched  booked
links    2026-08-30  party 6   0 matched  did not book
```

The six-person request on the 30th found nothing. That is a demand signal, and it belongs to the business. `POST /public/:slug/availability/search` is unauthenticated and still recorded; `GET /api/b/:slug/intent` reads it back.

Rules the code enforces

- Grants, not roles. No grant means denied. Explicit `never` beats everything.
- A capability marked sensitive parks in `awaiting_approval` and writes an approval row.
- No verifier, no claim. A capability with no `verify` gets a receipt of `unknown`.
- Receipts hash-chain per business; `verifyChain` catches an edit or a deletion.
- Uninstall revokes reach, not data.
- No per-channel columns. What an app shows is an observation with an authority rank.
- A manifest cannot declare `business_id`, cannot claim a capability outside its namespace, and a channel app cannot claim to carry a key it has no route for.
- Apps compose through events, not imports.
- Device writes are scripts. `agentSafe: false` keeps them out of every model's hands.
- The private surface is derived from grants, never from a fixed tool list.
- Tokens are stored hashed and scoped to one membership at one business.
- A provider must export what its slot's contract requires, checked at load.
- A harness is offered only what the acting person may already invoke.
- Nothing installs unsigned, and the signature covers the manifest hash.
- An import that fails any gate produces a rejection with the step and reason, never a partially trusted package.

HTTP surface

GET	/api/packages	what can be installed
GET	/api/capabilities	every registered verb
GET	/api/b/:slug/installs	what this business has installed
POST	/api/b/:slug/installs	install one
DELETE	/api/b/:slug/installs/:key	uninstall one
POST	/api/b/:slug/invoke	the only way to make anything happen
GET	/api/b/:slug/approvals	what is parked waiting on a human
POST	/api/b/:slug/approvals/:executionId	approve or reject
GET	/api/b/:slug/receipts	the ledger plus a chain check
GET	/api/b/:slug/drift/:key	canonical vs every app
GET	/api/b/:slug/workspaces	android instance and container
POST	/api/b/:slug/workspaces	provision
GET	/api/b/:slug/device/apps	apps on the device and login state
POST	/api/b/:slug/device/apps	install an app and log in
DELETE	/api/b/:slug/device/apps/:key	remove it
POST	/api/b/:slug/device/session	open the live screen
GET	/api/b/:slug/device/screen	what is on it now
GET	/api/b/:slug/sync	per app, per key, in sync or drift
POST	/api/b/:slug/sync/:key	force a fan-out
GET	/api/b/:slug/repairs	maps that stopped matching
GET	/api/automations	the automation marketplace
POST	/api/b/:slug/automations	publish one
POST	/api/b/:slug/automations/:key/install	install one by key
GET	/api/slots	every slot and its contract
GET	/api/b/:slug/providers	what is bound for this business
POST	/api/b/:slug/providers	swap a provider
POST	/api/b/:slug/think	route one task across model provide
GET	/api/b/:slug/routing	what was chosen, and what it fell b
POST	/api/b/:slug/harness	run an agent loop against a goal
GET	/api/b/:slug/harness	past runs
POST	/api/b/:slug/build	describe an app, get a plan
POST	/api/b/:slug/build/:planId/approve	deploy the plan as a package
POST	/api/b/:slug/build/:planId/reject	throw it away
GET	/api/import/steps	the import pipeline
POST	/api/b/:slug/import	bring a repo in
GET	/api/b/:slug/import	past import jobs
GET	/api/packages/:key/:version/signature	verify a package signature
POST	/api/b/:slug/tokens	issue a private-surface token
DELETE	/api/b/:slug/tokens/:id	revoke one
GET	/api/b/:slug/intent	what visitors searched for
GET	/p/:slug	public profile as JSON

GET	/l/:slug	the links page
GET	/embed/:slug/:section	one section as an embeddable fragme
POST	/public/:slug/availability/search	unauthenticated, and recorded
GET	/mcp/:slug	public surface
GET	/mcp/:slug/private	private surface (Bearer token)
GET	/mcp/:slug/:section	read one public resource

Providers shipped so far

package	fills	note
android_cloud	workspace.executor	persistent cloud instance
android_local_node	workspace.executor	same contract, on-premise hardware
hosted_models	model	declares models, cost, speed, strength
local_models	model	local: true , nothing leaves the workspace
loop_harness	harness	deterministic planner over allowed capabilities
composer	builder	intent to manifest

Apps shipped so far

package	declares	public section
business_profile	none; reads canonical	About, Hours, Contact
qr_menu	menu_item	Menu
availability	slot	Check Availability
forms	form , submission	Get in touch
crm	contact , deal	none
facebook / google_business / yelp	appmaps	none

Not built yet

The real ADB driver behind `src/drivers/android.js`. Login and credential capture on the device. Screen streaming and the control bridge — the session row and token exist, the transport does not. Screen reading by model instead of by appmap. Voice as an input. Real model endpoints — the providers declare and route correctly, but `complete()` simulates. A memory provider and a sandbox provider: the slots are declared, nothing fills them. Browser and desktop executors. Mesh. Dashboard and public UI; the API returns what they need. Consumer identity. Billing.